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VIRTUAL SPACE IMAGE PROVIDING DEVICE

Abstract

The virtual space image providing device includes a target-avatar extraction unit, a hint image generator, and a hint image superimposition unit. The target-avatar extraction unit extracts a second avatar as a target avatar when the second avatar exists in an area in a predetermined range outside a visual field seen by a first avatar in a virtual space image. The hint image generator generates a hint image that makes the first avatar feel that the second avatar is 10 outside the visual field. The hint image superimposition unit superimposes the hint image on an edge of the visual field.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of PCT Application No. PCT/JP2023/039017, filed on Oct. 30, 2023, and claims the priority of Japanese Patent Application No. 2022-187311, filed on Nov. 24, 2022, the entire contents of both of which are incorporated herein by reference.

BACKGROUND

[0002] The present disclosure relates to a virtual space image providing device.

[0003] Virtual space image providing devices that operate avatars corresponding to users in a virtual space are in widespread use.

SUMMARY

[0004] When a human looks in a certain direction, he or she may not only recognize something that exists within the range of his or her visual field, but also sense other human beings that exist in an area outside his or her field of view as a hint. Similarly, when an avatar looks in a certain direction in a virtual space, it is desired that the avatar be made to sense other avatars that exist in an area outside his or her field of view as a hint.

[0005] An aspect of one or more embodiments provides a virtual space image providing device including: a target-avatar extraction unit configured to extract a second avatar as a target avatar when the second avatar exists in a predetermined angular range outside a visual field which is a viewing angle of a first avatar in an virtual space image; a hint image generator configured to generate a hint image that makes the first avatar feel that the second avatar exists outside the visual field; and a hint image superimposition unit configured to superimpose the hint image on an edge of the visual field.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0006] FIG. 1 is a block diagram illustrating a virtual space image providing device according to one or more embodiments.

[0007] FIG. 2 is a block diagram illustrating a specific configuration example of a virtual space image generator in the virtual space image providing device according to one or more embodiments.

[0008] FIG. 3 is a conceptual diagram illustrating an example of a state in which the virtual space image generator in the virtual space image providing device according to one or more embodiments superimposes a hint image on a background image.

[0009] FIG. 4 is a conceptual diagram illustrating switching from a hint image to an avatar image.

[0010] FIG. 5 is a flowchart illustrating operation of the virtual space image providing device according to one or more embodiments.

DETAILED DESCRIPTION

[0011] A virtual space image providing device according to one or more embodiments will be described below with reference to the accompanying drawings. In FIG. 1, a virtual space image providing server 10 constitutes the virtual space image providing device according to one or more embodiments. The virtual space image providing server 10 and user terminals 30a to 30c are connected to a network 20. Any user terminal including the user terminals 30a to 30c and

unillustrated other user terminals will be referred to as the user terminal **30**. Typically, the network **20** is the Internet. The number of the user terminals **30** that are connected to the virtual space image providing server **10** via the network **20** is not limited.

[0012] The virtual space image providing server **10** includes a virtual space image generator **11**, an avatar image holding unit **12**, a visual field setting unit **13**, a background image setting unit **14**, an avatar position controller **15**, a direction indication unit **16**, and a communication unit **17**. Users Ura to Urc using the user terminals **30a** to **30c** wear head-mounted displays **40a** to **40c** on their heads and view virtual space images provided by the virtual space image providing server **10**. Users Ura to Urc and any user who uses an unillustrated user terminal **30** are referred to as user Ur. Any head-mounted display is referred to as the head-mounted display **40**.

[0013] The virtual space image generator **11** superimposes avatar images held in the avatar image holding unit **12** on a background image set in the background image setting unit **14** to generate a virtual space image. The virtual space image generator **11** superimposes the avatar images of avatars corresponding to the users Ur on the background image. Each user Ur can move the avatar by operating the user terminals **30**. The avatar position controller **15** controls the positions of the avatars in the background image according to the operation of the each user Ur. The virtual space image generator **11** positions the avatar at a predetermined position in the background image according to the control of the position of the avatar by the avatar position controller **15**.

[0014] The visual field setting unit **13** sets a visual field that the user Ur sees through the head-mounted display **40** of all virtual space images. A visual field is a predetermined angle range in a horizontal direction, and a predetermined angle range in a vertical direction. The visual field set in the visual field setting unit **13** may be fixed or variable depending on the head-mounted display **40** to be used. The visual field may be automatically set depending on the head-mounted display **40** to be used.

[0015] When the user Ur wearing the head-mounted display **40** moves his or her head to change the viewing direction, a signal that indicates orientation of the user Ur is supplied to the direction indication unit **16** via the network **20** and the communication unit **17**. In accordance with the signal that indicates the direction of the user Ur, the direction indication unit **16** supplies a direction indication signal to the virtual space image generator **11** to indicate the direction of the visual field to be extracted among all virtual space images. The virtual space image generator **11** extracts the image of the visual field area in an indicated direction from all the virtual space images, and supplies the extracted image to the communication unit **17**. The extracted image of the visual field area is transmitted to the head-mounted display **40** via the network **20** and the user terminals **30**.

[0016] In this way, the each user Ur can view an image of the visual field area in any direction among all the virtual space images by moving his or her head.

[0017] As illustrated in FIG. 2, the virtual space image generator **11** includes a visual field background image extraction unit **111**, an avatar image superimposition unit **112**, a target-avatar extraction unit **113**, a hint image generator **114**, and a hint image superimposition unit **115**. The visual field background image extraction unit **111** is supplied with a background image from the background image setting unit **14**, the visual field setting signal from the visual field setting unit **13**, and a direction indication signal from the direction indication unit **16**. The visual field background image extraction unit **111** extracts a background image of the visual field in an indicated direction.

[0018] The avatar image superimposition unit **112** is supplied with an avatar image from the avatar image holding unit **12**, and an avatar position control signal from the avatar position controller **15**. When an avatar to be superimposed exists in an extracted background image, the avatar image superimposition unit **112** superimposes an avatar image on the background image, and supplies it to the hint image superimposition unit **115**.

[0019] The target-avatar extraction unit **113** is supplied with a visual field setting signal, a direction indication signal, and an avatar position control signal. The target-avatar extraction unit **113**

extracts other avatar (second avatar) as a target avatar when the other avatar (second avatar) exists in a region of a predetermined range outside the visual field seen by the avatar (first avatar) corresponding to each user Ur . An avatar image and information indicating the target avatar extracted by the target-avatar extraction unit **113** are supplied to the hint image generator **114**. The hint image generator **114** generates a hint image which makes the avatar corresponding to each user Ur feel that another avatar exists outside the visual field, and supplies it to the hint image superimposition unit **115**.

[0020] The hint image superimposition unit **115** superimposes the hint image generated by the hint image generator **114** on the background image supplied from the avatar image superimposition unit **112**. At this point, the hint image superimposition unit **115** superimposes the hint image on an edge of the visual field.

[0021] The hint image generator **114** may generate a hint image independent of the avatar image of the target avatar. The hint image may be an image with a predetermined shape, such as a circle or an oval, for example. The hint image may be a relatively dark image such as a gray image.

[0022] It is preferable to generate a hint image based on the avatar image of a target avatar. As a first generating method, the hint image generator **114** may generate a hint image by lowering the brightness of the avatar image. As a second generating method, the hint image generator **114** may generate a hint image by changing the color of the avatar image. For example, the hint image generator **114** may change the color of the avatar image to a colder color or to a low saturation color.

[0023] As a third generating method, the hint image generator **114** may generate a hint image as a black and white image from the avatar image which is a color image. As a fourth generating method, the hint image generator **114** may generate a hint image by making the avatar image smaller. As a fifth generating method, the hint image generator **114** may generate a hint image by changing the shape of the avatar image. The hint image generator **114** may adopt at least one of the first to fifth generating methods, and may generate a hint image by combining any two or more of the first to fifth generating methods.

[0024] FIG. **3** conceptually illustrates a state in which the hint image generated by the hint image generator **114** is superimposed on the background image. An avatar $Av1$ corresponding to a certain user views a virtual space with an angle e as the visual field. The target-avatar extraction unit **113** sets an area in the range of angles θ_{exL} and θ_{exR} adjacent to the visual field and outside the visual field as expansion areas. The target-avatar extraction unit **113** determines whether other avatars exist in the expansion areas. In the example of FIG. **3**, an avatar $Av2$ exists in the area in the range of angle θ_{exR} , and an avatar $Av3$ exists in the area in the range of angle θ_{exL} .

[0025] As illustrated in FIG. **3**, the hint image generator **114** generates a hint image $Iav2$ based on the avatar image of the avatar $Av2$, and the hint image superimposition unit **115** superimposes the hint image $Iav2$ on an edge of the visual field. The hint image generator **114** generates a hint image $Iav3$ based on the avatar image of the avatar $Av3$, and the hint image superimposition unit **115** superimposes the hint image $Iav2$ on an edge of the visual field. It is preferable that the hint image superimposition unit **115** superimposes the hint image $Iav2$ in a location at the same distance as the distance from the avatar $Av1$ to the avatar $Av2$, and superimposes the hint image $Iav3$ on a location at the same distance as the distance from the avatar $Av1$ to the avatar $Av3$. That is, the hint images $Iav2$ and $Iav3$ may be arranged at locations where the avatars $Av2$ and $Av3$ are circumferentially moved to the edges of the visual field.

[0026] According to the virtual space image illustrated in FIG. **3**, the avatar $Av1$ (the user Ur corresponding to the avatar $Av1$) can sense other avatars $Av2$ and $Av3$ which exist in the area outside the field of view of the avatar $Av1$ as a hint. The hint images $Iav2$ and $Iav3$ are images different from the avatar images of the avatars $Av2$ and $Av3$, so that there is little possibility that the avatar $Av1$ (the user Ur corresponding to the avatar $Av1$) recognizes that the avatars $Av2$ and $Av3$ exist in the visual field.

[0027] It is preferable that the hint image generator **114** generates a hint image that is larger as the distance from the avatar Av1 to the avatar in the extended area is shorter, and that is smaller as the distance from the avatar Av1 to the avatar in the extended area is longer. It is preferable that the hint image generator **114** generates a hint image that is larger as the avatar in the extended area is closer to the visual field, and generates a hint image that is smaller as the distance is farther away from the visual field.

[0028] In FIG. 3, when the user corresponding to the avatar Av1 rotates his or her face to the right, the visual field of angle ϵ rotates to the right, and the avatar Av2 enters the visual field, so that the user sees the avatar Av2. At this point, it is preferable that the hint image Iav2 and the avatar Av2 are continuously displayed as switching from the hint image Iav2 to the avatar Av2. FIG. 4 is conceptually illustrates a state in which the image Iav2 is switched to the avatar Av2. The image Iav2 successively changes to an intermediate image Iav2' between the image Iav2 and the avatar Av2, and then successively changes to the avatar Av2.

[0029] The processing executed by the virtual space image generator **11** will be described with reference to the flowchart illustrated in FIG. 5. In FIG. 5, when the virtual space image providing server **10** starts operating, and the processing of the virtual space image generator **11** starts, the virtual space image generator **11** extracts the background image of the visual field from the background image based on the visual field setting signal and the direction indication signal in step S1. The virtual space image generator **11** determines whether the avatar exists in the background image of the visual field in step S2. If the avatar does not exist in the background image of the visual field (NO), the virtual space image generator **11** shifts the processing to step S4.

[0030] If the avatar exists in the background image of the visual field in step S2 (YES), the virtual space image generator **11** superimposes the avatar image on the background image of the visual field in step S3 and shifts the processing to step S4. The virtual space image generator **11** determines whether the avatar exists in the extended area outside the visual field in step S4. If the avatar does not exist in the extended area outside the visual field (NO), the virtual space image generator **11** shifts the processing to step S6.

[0031] If the avatar exists in the extended area outside the visual field (YES) in step S4, the virtual space image generator **11** superimposes the image of the avatar on an edge of the visual field in step S5, and shifts the processing to step S6. In step S6, the virtual space image generator **11** determines whether the direction in which the user faces has changed. If the direction in which the user faces has not changed (NO), the virtual space image generator **11** shifts the processing to step S7. If the direction in which the user faces has changed (YES), the virtual space image generator **11** returns the processing to step S1, and repeats the processing from step S1.

[0032] In step S7, the virtual space image generator **11** determines whether operation should be terminated by terminating operation of the virtual space image providing server **10** or the like. If the operation is not terminated (NO), the virtual space image generator **11** repeats the process from step S2. If the operation is terminated (YES), the virtual space image generator **11** terminates the processing.

[0033] As described above, in accordance with the virtual space image providing device according to one or more embodiments, it is possible to make the avatar feel a hint of another avatar that exists in an area outside the field of view of the avatar.

[0034] The present invention is not limited to one or more embodiments described above, and various modifications can be made without departing from the scope of the present invention. The virtual space image providing server **10** can be configured with a computing device. A central processing unit of the computing device may execute the operation of the virtual space image providing server **10** including the operation of the virtual space image generator **11** described above by executing a computer program (virtual space image providing program).

Claims

1. A virtual space image providing device comprising: a target-avatar extraction unit configured to extract a second avatar as a target avatar when the second avatar exists in a predetermined angular range outside a visual field which is a viewing angle of a first avatar in an virtual space image; a hint image generator configured to generate a hint image that makes the first avatar feel that the second avatar exists outside the visual field; and a hint image superimposition unit configured to superimpose the hint image on an edge of the visual field.
 2. The virtual space image providing device according to claim 1, wherein the hint image generator is configured to generate the hint image based on an avatar image of the second avatar.
 3. The virtual space image providing device according to claim 1, wherein the hint image generator is configured to generate a hint image that is larger as a distance from the first avatar to the second avatar is shorter, and to generate a hint image that is smaller as the distance from the first avatar to the second avatar is longer.
 4. The virtual space image providing device according to claim 1, wherein the hint image generator is configured to generate a hint image that is larger as the position of the second avatar approaches the visual field, and to generate a hint image that is smaller as the position of the second avatar moves away from the visual field.
 5. A virtual space image providing device according to claim 2, wherein the hint image generator is configured to generate the hint image by at least one of a first generating method of reducing luminance of the avatar image of the second avatar, a second generating method of changing color of the avatar image, a third generating method of making the avatar image which is a color image a black and white image, a fourth generating method of making the avatar image smaller, and a fifth generating method of changing a shape of the avatar image.
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